## Installing System Software

If you're using a virtual machine, the commands that follow assume you've logged in to it with `vagrant ssh`. Otherwise, we'll assume you're logged in as a user who can execute `sudo`. Also note that in this case, you may have some software installed already. If you experience problems, you might want to update that software to the latest versions.

Many Ruby libraries are actually wrappers for C libraries, and when you install them, your system will try to build those libraries or build native connectors. This is the main reason we need certain software installed before we get to Ruby. First, refresh the list of packages available for your operating system:

```
$ sudo apt-get update
```

That will produce a large amount of output. Once that's done, we'll install several different libraries and tools. What this will look like on Ubuntu and most Debian-based Linuxes is as follows:

```
$ sudo apt-get install -y \
    build-essential \
    git \
    libsqlite3-dev \
    redis \
    ruby-dev \
    tzdata
```

For Centos Stream and most RedHat-based Linuxes, the following will get you started:

```
$ sudo yum install -y \
    gcc \
    git \
    redis \
    ruby-devel \
    sqlite-devel \
    which
$ sudo yum reinstall -y tzdata
```

Now we can install Ruby and Rails!